

# CivicRoute Service Request Triage

Project Report

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## Abstract

This report presents an interpretable text-classification project for first-level public service-request triage. I prepared 36,805 labelled and deduplicated requests from 175,784 source records, trained a class-balanced TF-IDF logistic-regression pipeline, and added confidence-based manual review.

A random holdout reached 0.995 macro F1. A harder holdout that excluded complete category codes from training reached 0.974 macro F1 on 12,789 requests. Five experiments examine generalisation, confidence coverage, label imbalance, class-level performance, and accepted-route accuracy.

**Keywords:** text classification; service requests; grouped validation; confidence threshold; human review

## Report structure

The report presents the problem, source data, preparation method, evaluation design, five evidence-based experiments, limitations, reproducibility details, and conclusion.

## **1. Project overview and problem statement**

CivicRoute reads a written public service request and suggests one of eight service queues. I built it because free-text requests and structured administrative queues do not naturally match, which can create transfers and delay a first response.

The system is an interpretable triage experiment. It does not file a complaint, validate the facts, decide priority or legal rights, or replace an official portal. Low-confidence and very short messages are retained for manual review.

## 2. Source, privacy and preparation

The source is a 175,784-record Government of India complaint corpus published on Kaggle with an MIT licence and a category mapping workbook. The source system uses the formal word grievance; the project interface uses service request because it is clearer.

I map category codes to organisations, select eight well-represented queues, remove insufficient text, clean repeated redaction markers, and remove exact duplicate request text. The prepared table contains 36,805 rows. Generated text is excluded from Git because de-identified requests can still contain details entered by a citizen.

| Service queue         | Prepared rows |
|-----------------------|---------------|
| Labour and Employment | 9,458         |
| Financial Services    | 6,272         |
| Direct Taxes          | 5,978         |
| Agriculture           | 5,220         |
| Telecommunications    | 3,622         |
| Postal Services       | 3,058         |
| Railways              | 2,048         |
| Road Transport        | 1,149         |

### **3. Model and evaluation design**

The pipeline uses TF-IDF unigrams and bigrams with class-balanced logistic regression. The saved pipeline contains both text transformation and classification. Predictions below 0.58 confidence are not routed automatically.

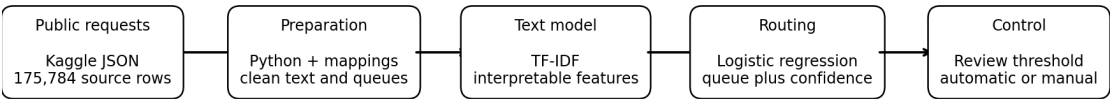
A random stratified holdout measures new rows from familiar administrative categories. A harder category-group holdout keeps complete category codes out of training and is the headline generalisation test. This reduces the benefit of category-specific language shared between training and test rows.

## 4. End-to-end routing architecture

The routing architecture begins with the public no-PII request file and declared category mapping. Python prepares text and service ownership, TF-IDF creates interpretable text features, and logistic regression returns one queue with a confidence score.

The confidence policy is a separate control stage. A low score does not become a forced prediction; it becomes a manual-review decision. This keeps model uncertainty visible in both the evaluation and the runtime interface.

CivicRoute end-to-end routing architecture



Architecture diagram. CivicRoute preparation, modelling and review-control stages.

**Stage design:** The diagram distinguishes feature extraction, classification and the review decision instead of treating them as one model call.

**Technology:** Python, TF-IDF and logistic regression keep the route evidence inspectable and reproducible.

**Control:** Requests below the declared confidence threshold are retained for human review rather than silently assigned.

## 5. Automated routing test execution

I ran the current repository test suite after revising the report. Four tests passed. The suite checks a clear railway request, strict-threshold review behaviour and required repository assets.

Eight deprecation warnings were emitted by joblib when loading arrays under NumPy 2.5. They are recorded because warnings should not be hidden, but they did not change the passing result or the evaluated predictions.

### Actual routing test execution

```
$ .venv/bin/pytest -q
tests/test_router.py .... [100%]

4 passed, 8 dependency warnings in 2.06s

Validated: clear-route inference, confidence fallback,
repository assets and manual-review enforcement.
Warnings originate from joblib with NumPy 2.5.
```

*Test execution evidence. Actual CivicRoute pytest result with dependency-warning disclosure.*

**Execution:** Four tests passed and no test failed.

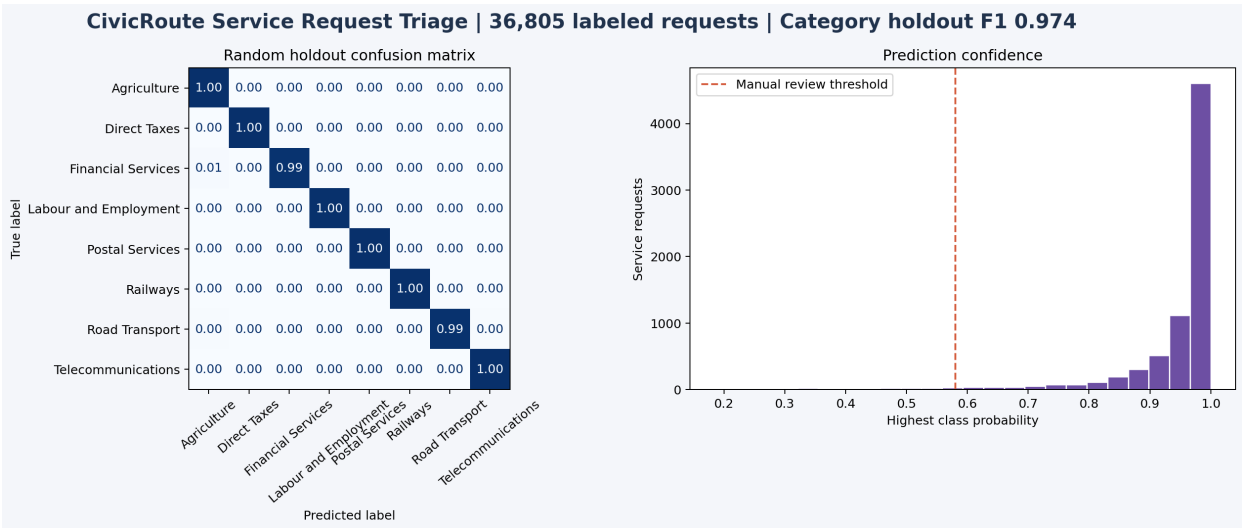
**Negative scenario:** A deliberately strict threshold forces manual review, verifying the safe fallback path.

**Maintenance item:** The joblib and NumPy compatibility warning should be removed in a future dependency update.

## 6. Routing evaluation dashboard

The evaluation dashboard combines both holdouts, automatic coverage, confidence review and queue performance. This is important because a high accuracy value alone can hide a model that routes only easy requests or sees the same category structure during training.

The category-group holdout remains the headline scenario. It measures how the model behaves when complete category codes are absent from training and therefore provides a stricter view of generalisation.



Evaluation dashboard. CivicRoute random and unseen-category scenarios.

**Question:** Can the system preserve accuracy while retaining uncertain requests for review?

**Observed result:** The category-group holdout reaches 0.974 macro F1 and sends 15.2 percent of requests to review.

**Interpretation:** Coverage and accuracy are read together so the confidence threshold is not credited for predictions it declined to automate.

## 7. Experiment 1: holdout comparison

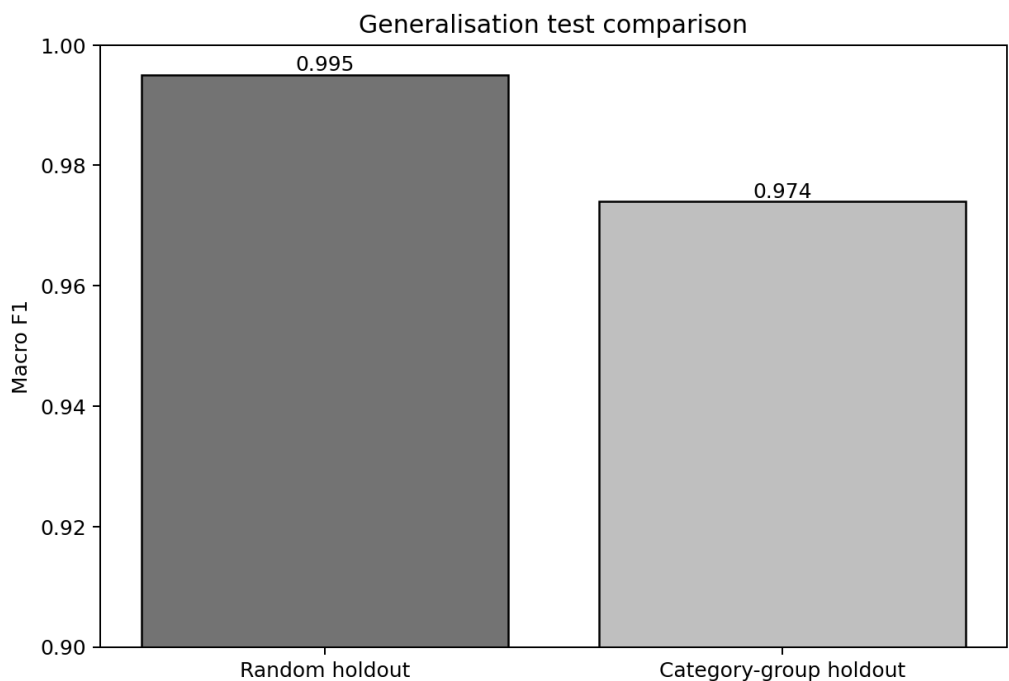


Figure 1. Macro F1 under random and category-group holdouts.

**What I tested:** Whether the model still works when complete administrative category codes are unseen during training.

**What the graph shows:** Macro F1 falls from 0.995 on the random holdout to 0.974 on the harder split.

**Conclusion:** The random split is optimistic; the category-group result is the more credible headline measure.

## 8. Experiment 2: confidence review

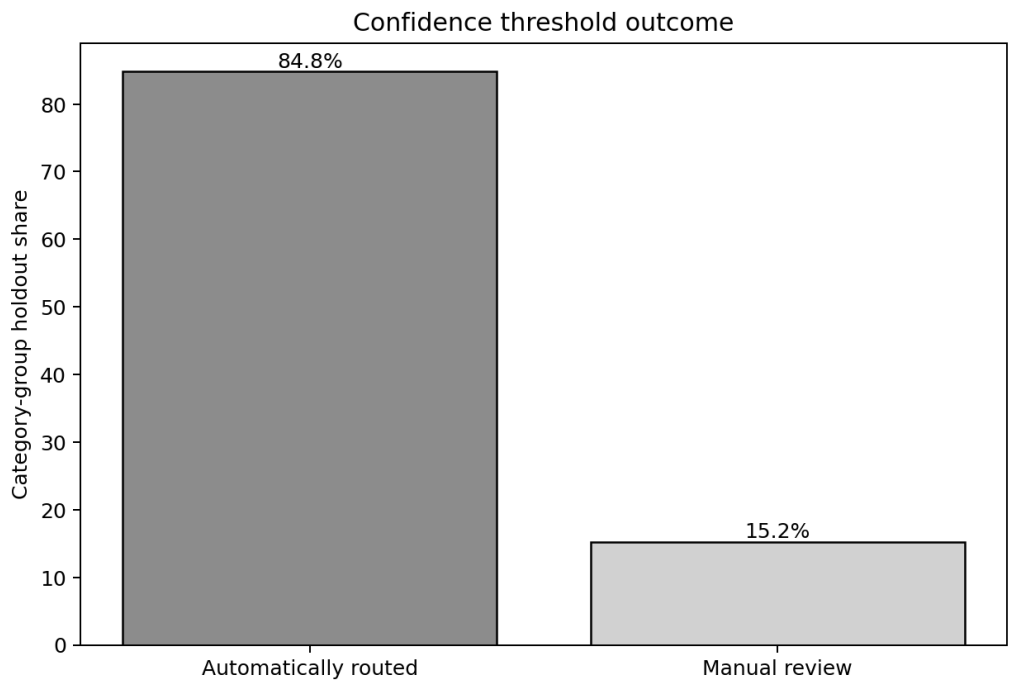


Figure 2. Automatic routing and manual-review shares on the category-group holdout.

**What I tested:** How much work is accepted automatically at the 0.58 confidence threshold.

**What the graph shows:** 84.8 percent is automatically routed and 15.2 percent is retained for review.

**Conclusion:** The model exposes uncertainty rather than forcing every request into a queue.

### 9. Experiment 3: queue distribution

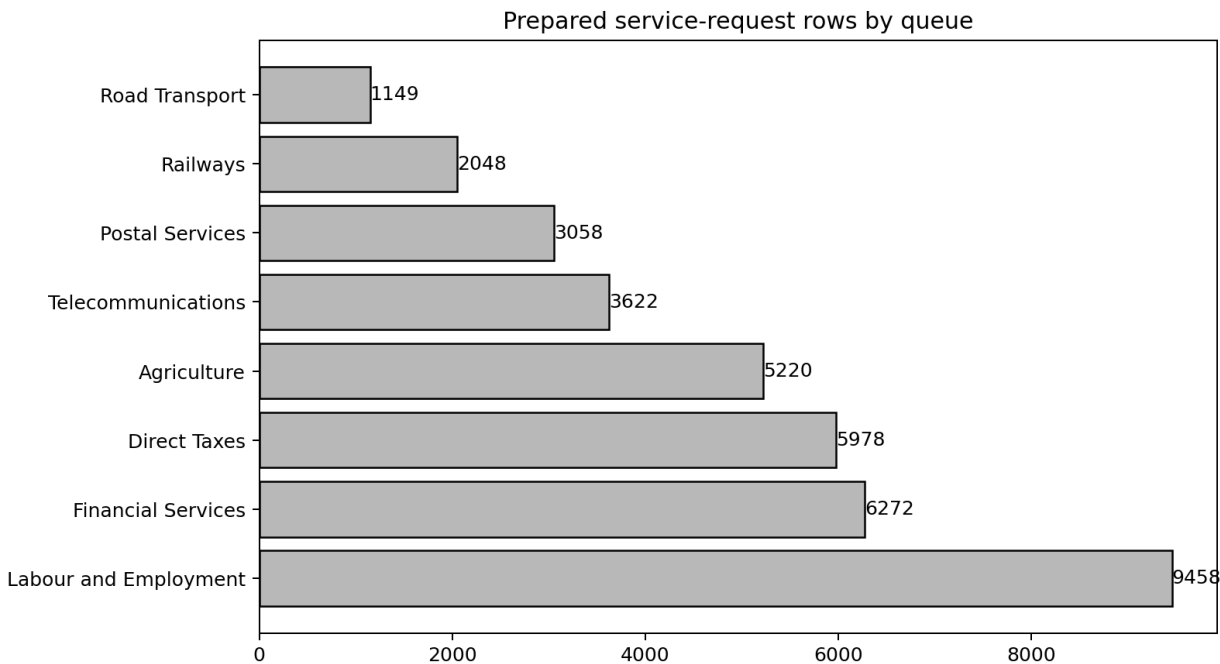


Figure 3. Prepared examples across the eight service queues.

**What I tested:** Whether the training labels are balanced across queues.

**What the graph shows:** Labour and Employment has more than eight times the rows of Road Transport.

**Conclusion:** Class balancing and macro F1 are necessary; accuracy alone would understate minority-queue risk.

## 10. Experiment 4: class-level F1

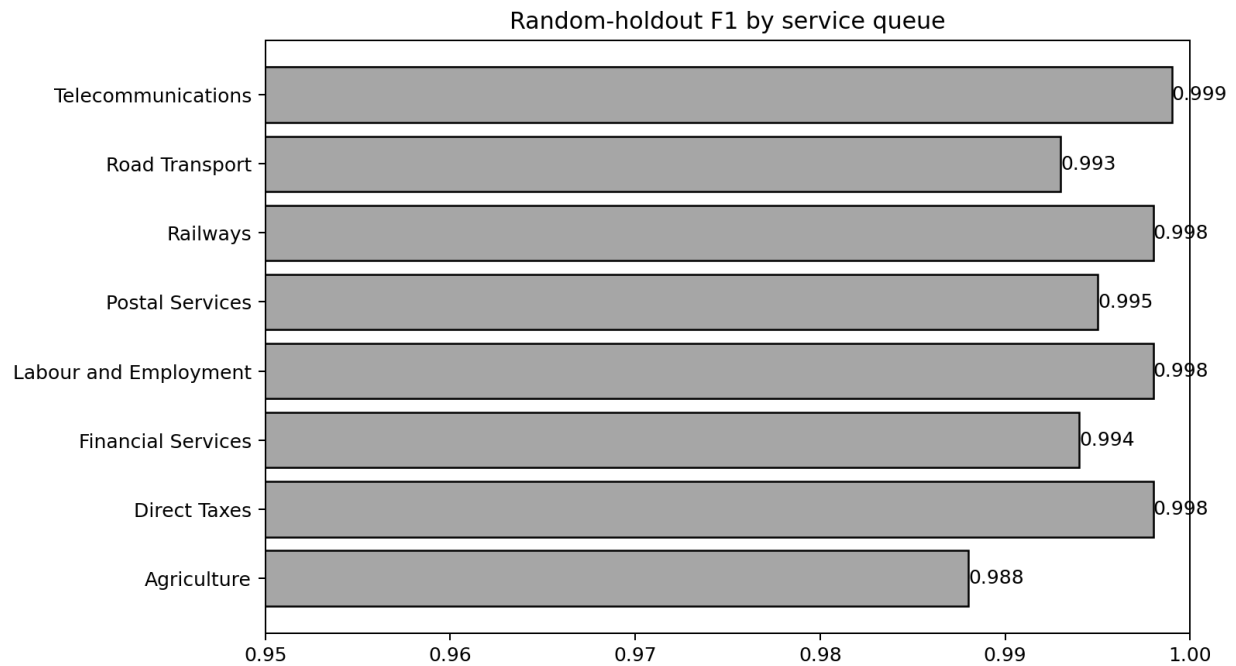


Figure 4. Random-holdout F1 for each service queue.

**What I tested:** Whether one aggregate score hides a weak service queue.

**What the graph shows:** All eight random-holdout F1 values are high, with Agriculture the lowest at 0.988.

**Conclusion:** Class-level performance is strong on familiar categories, but this does not replace the harder grouped evaluation.

## 11. Experiment 5: confidence and accuracy

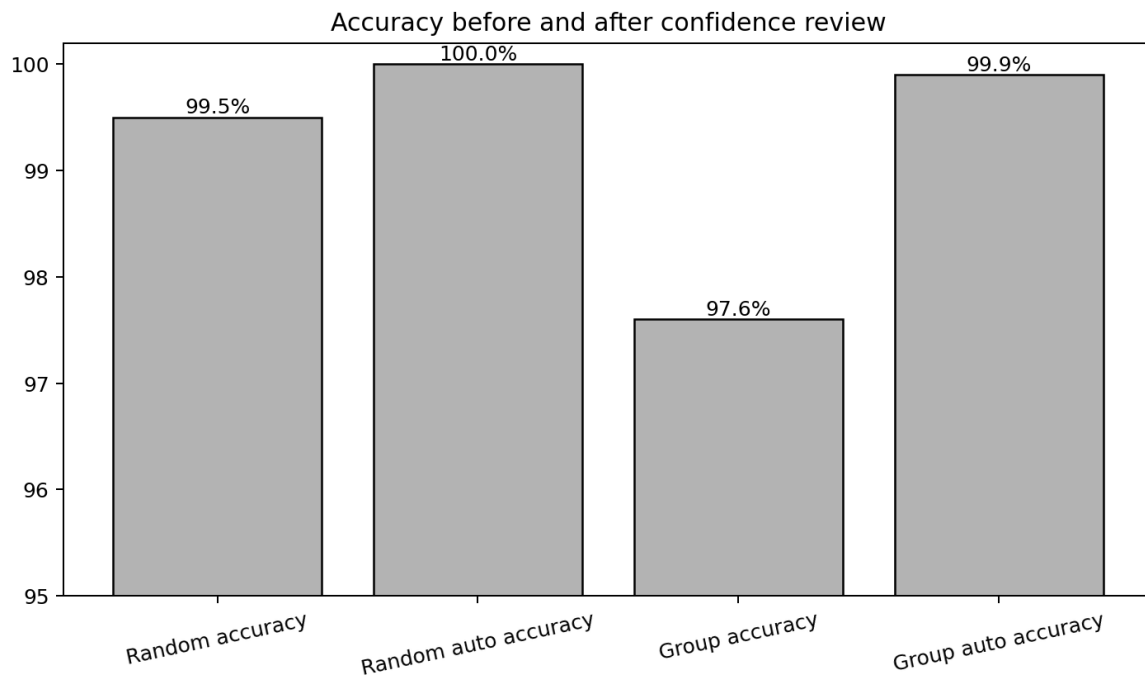


Figure 5. Overall and accepted-route accuracy under both holdouts.

**What I tested:** Whether manual review improves the reliability of automatically accepted suggestions.

**What the graph shows:** Accepted-route accuracy reaches 99.9 percent on the grouped holdout while coverage is 84.8 percent.

**Conclusion:** The threshold creates a useful accuracy-coverage trade-off, but it cannot detect every confidently wrong request.

## 12. Results and interpretation

The random holdout contains 7,361 rows and reaches 99.5% accuracy with macro F1 0.995. The category-group holdout contains 12,789 rows from 90 unseen category codes and reaches 97.6% accuracy with macro F1 0.974.

At the selected threshold, grouped automatic coverage is 84.8 percent and accepted-route accuracy is 99.9 percent in this snapshot. This result is not a claim about every future organisation, language, emergency request, or taxonomy.

### **13. Limitations, governance and reproducibility**

The corpus is historical and English-dominant. Administrative taxonomies change, similar language can occur across queues, and compound requests may require multiple owners. The system does not detect urgency, malicious content, or sensitive information reliably.

The repository versions acquisition, preparation, both evaluation splits, threshold behaviour, tests, five figures, the model card, and this report. Future work should add time based validation, independently written and multilingual requests, queue specific thresholds, drift monitoring, emergency detection, and a secure analyst review interface.

## 14. Conclusion

CivicRoute provides an interpretable and measurable approach to service request triage. The central result is not the near perfect random split; it is the deliberate use of a harder unseen category split and a visible manual review path. The project shows how NLP evaluation, class imbalance, confidence thresholds, privacy boundaries, and governance fit together in a bounded system.